University of Stirling

ITNPBD2 Python for Data Science

Assignment Autumn 2026

A Consultancy Job for JCPenney

This notebook forms the assignment instructions and submission document of the assignment for ITNPBD2. Read the instructions carefully and enter code into the cells as indicated.

You will need these files, which were in the Zip file you can download from the course webpage. The data in these files describes products that have been sold by the American retail giant, JCPenney, and reviews by customers who bought them. Note that the product data is real, but the customer data is synthetic.

- jcpenney_products.json - Details about the products for sale on the JCPenney website
- jcpenney_customers.json - Descriptions of the customer who have bought products and a list of the products they have reviewed
- reviews.csv - Individual reviews for products along with a user ID and a rating out of 5

The data sets contain structured, semi-structured and unstructured (text) data and your submission should show that you can work with all three.

The datasets are linked by product and customer IDs and your code should show how you joined data across different sources.

The Assignment

You should complete the assignement described below by adding code and markdown cells to this notebook to produce a coherent report on the data you have been given. Use markdown cells to pose questions about the data, then write code to answer the question, then use a markdown cell to comment on what the results of running the code tell you. You should also make sure that your code has informative comments embedded within it.

Make use of different coding skills such as defining and calling functions and generators, using imported modules, and storing data in suitable data structures.

Your task is to use the data provided to demonstrate your Python data manipulation skills in the form of a descriptive analysis of the data. By descriptive, we mean that you do not need to perform statistical tests or build machine learning models - simply present insights from the data in the form of descriptive statistics and visualisations.

With the text data, you might consider performing sentiment analysis, creating word clouds, or analysing vocabulary. There are location variables in the data, which you could use in visualisations or to describe geographic differences.

You should acquire additional data from at least one other source too - either download it directly, access it using `requests` or an API, or scrape it if you know how.

Produce working code to implement your ideas in as many cells as you need below. There is no single right answer, the aim is to simply show you are competent in using python for data analysis. Exactly how you do that is up to you.

For a distinction class grade, this must show originality, creative thinking, and insights beyond what you've been taught directly on the module.

Structure

You may structure the project how you wish, but here is a suggested guideline to help you organise your work:

1. Data Exploration - Explore the data and show you understand its structure and relations
2. Data Validation - Check the quality of the data. Is it complete? Are there obvious errors?
3. Data Visualisation - Gain an overall understanding of the data with visualisations
4. Data Analysis - Set some questions and use the data to answer them (you can use visualisations to do this)
5. Data Augmentation - Add new data from another source to bring new insights to the data you already have

Use the notebook to create a professional looking report. Use markdown cells to describe the questions you set out to answer via the data, then show the code and output that generates the answer, then explain the answer in a further markdown cell. Try to avoid producing a series of unrelated computations.

Completing the Assignment

Rename this file to be xxxxxx_BD2 where xxxxxx is your student number, then type your code and narrative description into the boxes provided. Add as many code and markdown cells as you need. The cells should contain:

- **Text narrative describing what you did with the data**
- **The code that performs the task you have described**
- **Comments that explain your code**

Marking Scheme

The assessment will be marked against the university Common Marking Scheme (CMS)

Here is a summary of what you need to achieve to gain a grade in the major grade bands:

Grade	Requirement
Fail	You will fail if your code does not run or does not achieve even the basics of the task. You may also fail if you submit code without either comments or a text explanation of what the code does.
Pass	To pass, you must submit sufficient working code to show that you have mastered the basics of the task, even if not everything works completely. You must include some justifications for your choice of methods, but without mentioning alternatives.
Merit	For a merit, your code must be mostly correct, with only small problems or parts missing, and your comments must be useful rather than simply re-stating the code in English. Most choices for methods and structures should be explained and alternatives mentioned.

| Distinction | For a distinction, your code must be working, correct, and well commented and shows an appreciation of style, efficiency and reliability. All choices for methods and structures are concisely justified and alternatives are considered. For a distinction, your work should be good enough to present to executives at a company and show originality, creative thinking, and insights beyond what you've been taught directly on the module.

The full details of the CMS can be found here

https://www.stir.ac.uk/about/professional-services/student-academic-and-corporate-services/academic-registry/academic-policy-and-practice/quality-handbook/assessment-policy-and-procedure/appendix-2-postgraduate-common-marking-scheme/

Note that this means there are not certain numbers of marks allocated to each stage of the assignment. Your grade will reflect how well your solutions and comments demonstrate that you have achieved the learning outcomes of the task.

Submission

When you have completed your notebook, save it as xxxxx.pdf where xxxxx is your student number. Then upload the file to the submission page on Canvas. There are several ways to convert a notebook to a PDF, depending on the software you are using. You could try Save As or Export as or print to PDF. If you really cannot export to PDF, then upload the .ipynb file instead.

Late penalties will apply at a rate of three marks per day, up to a maximum of 7 days. After 7 days you will be given a mark of 0. Extensions will be considered under acceptable circumstances outside your control.

Academic Integrity

This is an individual assignment, and so all submitted work must be fully your own work.

The University of Stirling is committed to protecting the quality and standards of its awards. Consequently, the University seeks to promote and nurture academic integrity, support staff academic integrity, and support students to understand and develop good academic skills that facilitate academic integrity.

In addition, the University deals decisively with all forms of Academic Misconduct.

Where a student does not act with academic integrity, their work or behaviour may demonstrate Poor Academic Practice or it may represent Academic Misconduct.

AI Usage

You should read the University guidance on the use of generative Artificial Intelligence (AI), for example ChatGPT, at the University. Allowed use of AI assistance in this assignment is level 2: AI can be used in the assessment for brainstorming, creating structures and generating ideas for improving work. No AI content is allowed in the final submission and you must not upload the questions or the data to an AI app.

Poor Academic Practice

Poor Academic Practice is defined as: "The submission of any type of assessment with a lack of referencing or inadequate referencing which does not effectively acknowledge the origin of words, ideas, images, tables, diagrams, maps, code, sound and any other sources used in the assessment."

Academic Misconduct

Academic Misconduct is defined as: "any act or attempted act that does not demonstrate academic integrity and that may result in creating an unfair academic advantage for you or another person, or an academic disadvantage for any other member or member of the academic community."

Plagiarism is presenting somebody else's work as your own. Plagiarism is a form of academic misconduct and is taken very seriously by the University. Students found to have plagiarised work can have marks deducted and, in serious cases, even be expelled from the University. Do not submit any work that is not entirely your own. Do not collaborate with or get help from anybody else with this assignment.

The University of Stirling's full policy on Academic Integrity can be found at:

https://www.stir.ac.uk/about/professional-services/student-academic-and-corporate-services/academic-registry/policy-and-procedure/

Add Code and Markdown Here

Here is the first markdown cell and below is the first code cell. Use these and then add as many more as you need

3547468 – ITNPBD2 Python for Data Science

A Consultancy Job for JCPenney – Data Analysis Report

This notebook presents my analysis of the JCPenney datasets as part of the ITNPBD2 Python for Data Science assignment. The main aim of this project is to explore the data and understand what it can tell us about the company's products, customers and customer reviews. For this project, I am working with three datasets:

- **jcpenney_products.json** – contains information about products available on the JCPenney website.
- **jcpenney_customers.json** – contains customer profiles and details of the reviews they have written.
- **reviews.csv** – contains individual product reviews, including customer ratings and written feedback.

These datasets contain different types of information, but they are connected through product IDs and customer IDs. This allows me to combine the information and explore it in more detail.Throughout this project, I will focus on:

- Exploring the datasets to understand their contents and how they are connected.
- Checking the data for missing values, duplicates and any inconsistencies.
- Creating charts and visualisations to identify patterns in product ratings and customer behaviour.
- Investigating questions about the products and customers using Python.
- Finding additional information from external sources to support my analysis.
- Explaining the results and highlighting any useful findings.

The report is organised into different sections, starting with exploring the data and moving towards more detailed analysis. I will use Python throughout the project to examine the datasets, present the results and identify findings that could help JCPenney better understand its customers and products.

Section 1: Data Exploration

I begin this section by taking a closer look at the three datasets provided for the assignment. Before starting the main analysis, I want to understand what information each file contains and how the data is organised. I will also look at how the three datasets are linked together through customer IDs and product IDs. To get a better understanding of the data, I will:

- Look at the first few rows of each dataset to see what the data looks like.
- Check how many records are available in each file.
- Examine the column names and data types.
- Find out how the customer, product and review information is connected.

This will help me become familiar with the data and give me a better idea of how to approach the next stages of the assignment.

In [107…
```
!pip install textblob
```

```
Requirement already satisfied: textblob in C:\Users\shahb\anaconda3\Lib\site-packages (0.20.1)
Requirement already satisfied: nltk>=3.9 in C:\Users\shahb\anaconda3\Lib\site-packages (from textblob) (3.10.3)
Requirement already satisfied: defusedxml in C:\Users\shahb\anaconda3\Lib\site-packages (from nltk>=3.9->textblob) (0.7.1)
Requirement already satisfied: click in C:\Users\shahb\anaconda3\Lib\site-packages (from nltk>=3.9->textblob) (8.4.1)
Requirement already satisfied: joblib in C:\Users\shahb\anaconda3\Lib\site-packages (from nltk>=3.9->textblob) (1.6.0)
Requirement already satisfied: regex>=2021.8.3 in C:\Users\shahb\anaconda3\Lib\site-packages (from nltk>=3.9->textblob) (2026.7.19)
Requirement already satisfied: tqdm in C:\Users\shahb\anaconda3\Lib\site-packages (from nltk>=3.9->textblob) (4.68.2)
Requirement already satisfied: colorama in C:\Users\shahb\anaconda3\Lib\site-packages (from click->nltk>=3.9->textblob) (0.4.6)
Requirement already satisfied: cloudpickle>=3.0 in C:\Users\shahb\anaconda3\Lib\site-packages (from joblib->nltk>=3.9->textblob) (3.1.2)
```

In [108…
```
!python -m textblob.download_corpora
```

```
Finished.
[nltk_data] Downloading package brown to
[nltk_data]     C:\Users\shahb\AppData\Roaming\nltk_data...
[nltk_data]   Package brown is already up-to-date!
[nltk_data] Downloading package punkt_tab to
[nltk_data]     C:\Users\shahb\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt_tab is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\shahb\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger_eng to
[nltk_data]     C:\Users\shahb\AppData\Roaming\nltk_data...
[nltk_data]   Package averaged_perceptron_tagger_eng is already up-to-
[nltk_data]       date!
[nltk_data] Downloading package conll2000 to
[nltk_data]     C:\Users\shahb\AppData\Roaming\nltk_data...
[nltk_data]   Package conll2000 is already up-to-date!
[nltk_data] Downloading package movie_reviews to
[nltk_data]     C:\Users\shahb\AppData\Roaming\nltk_data...
[nltk_data]   Package movie_reviews is already up-to-date!
```

In [109…
```
# Section 1: Data Exploration

import json
import pandas as pd

# Read the customer data
customers = []
with open("jcpenney_customers.json", "r") as file:
    for line in file:
        customers.append(json.loads(line))

# Read the product data
products = []
with open("jcpenney_products.json", "r") as file:
    for line in file:
        products.append(json.loads(line))

# Read the reviews data
reviews = pd.read_csv("reviews.csv")

# Convert the customer and product data into DataFrames
customers_df = pd.DataFrame(customers)
products_df = pd.DataFrame(products)

# Display the first few rows of each dataset
print("Customer Data:")
display(customers_df.head())

print("Product Data:")
display(products_df.head())

print("Review Data:")
display(reviews.head())
```

Customer Data:

	Username	DOB	State	Reviewed
0	bkpn1412	31.07.1983	Oregon	[cea76118f6a9110a893de2b7654319c0]
1	gqjs4414	27.07.1998	Massachusetts	[fa04fe6c0dd5189f54fe600838da43d3]
2	eehe1434	08.08.1950	Idaho	[]
3	hkxj1334	03.08.1969	Florida	[f129b1803f447c2b1ce43508fb822810, 3b0c9bc0be6...
4	jjbd1412	26.07.2001	Georgia	[]

Product Data:

	uniq_id	sku	name_title	description	list_price	sale_price	category	category_tree	average_product_rating	
0	b6c0b6bea69c722939585baeac73c13d	pp5006380337	Alfred Dunner® Essential Pull On Capri Pant	You'll return to our Alfred Dunner pull-on cap...	41.09	24.16	alfred dunner	jcpenney|women|alfred dunner	2.625	http:
1	93e5272c51d8cce02597e3ce67b7ad0a	pp5006380337	Alfred Dunner® Essential Pull On Capri Pant	You'll return to our Alfred Dunner pull-on cap...	41.09	24.16	alfred dunner	jcpenney|women|alfred dunner	3.000	http:
2	013e320f2f2ec0cf5b3ff5418d688528	pp5006380337	Alfred Dunner® Essential Pull On Capri Pant	You'll return to our Alfred Dunner pull-on cap...	41.09	24.16	view all	jcpenney|women|view all	2.625	http:
3	505e6633d81f2cb7400c0cfa0394c427	pp5006380337	Alfred Dunner® Essential Pull On Capri Pant	You'll return to our Alfred Dunner pull-on cap...	41.09	24.16	view all	jcpenney|women|view all	3.500	http:
4	d969a8542122e1331e304b09f81a83f6	pp5006380337	Alfred Dunner® Essential Pull On Capri Pant	You'll return to our Alfred Dunner pull-on cap...	41.09	24.16	view all	jcpenney|women|view all	3.125	http:

Review Data:

	Uniq_id	Username	Score	Review
0	b6c0b6bea69c722939585baeac73c13d	fsdv4141	4	You never have to worry about the fit...Alfred...
1	b6c0b6bea69c722939585baeac73c13d	krpz1113	5	Good quality fabric. Perfect fit. Washed very ...
2	b6c0b6bea69c722939585baeac73c13d	mbmg3241	4	I do not normally wear pants or capris that ha...
3	b6c0b6bea69c722939585baeac73c13d	zeqg1222	5	I love these capris! They fit true to size and...
4	b6c0b6bea69c722939585baeac73c13d	nvfn3212	5	This product is very comfortable and the fabri...

Initial Exploration Summary

After displaying the first few rows, I can see that each dataset contains different types of information. The customer file provides details such as usernames, dates of birth, states and review IDs. The product file contains product IDs and other details about the products, while the reviews file includes review IDs, product IDs, ratings and written customer feedback. I can also see that the datasets share some common IDs, which will help me connect the information between them. This will be useful later when I combine the datasets to look more closely at customer reviews, product ratings and customer behaviour.

Question: What are the main columns and data types in each dataset?

Next, I will check the columns and data types in all three datasets to get a better understanding of the information they contain. This will help me see what type of data I am working with, identify any differences between the datasets and check whether the data is in a suitable format for further analysis.

In [110…

```
# Check the structure and data types of each dataset

print("Customer Dataset:")
customers_df.info()

print("\nProduct Dataset:")
products_df.info()

print("\nReview Dataset:")
reviews.info()
```

```
Customer Dataset:
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Username  5000 non-null   str
 1   DOB       5000 non-null   str
 2   State     5000 non-null   str
 3   Reviewed  5000 non-null   object
dtypes: object(1), str(3)
memory usage: 290.4+ KB

Product Dataset:
<class 'pandas.DataFrame'>
RangeIndex: 7982 entries, 0 to 7981
Data columns (total 15 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   uniq_id                 7982 non-null   str
 1   sku                     7982 non-null   str
 2   name_title              7982 non-null   str
 3   description             7982 non-null   str
 4   list_price              7982 non-null   str
 5   sale_price              7982 non-null   str
 6   category                7982 non-null   str
 7   category_tree           7982 non-null   str
 8   average_product_rating  7982 non-null   float64
 9   product_url             7982 non-null   str
 10  product_image_urls      7982 non-null   str
 11  brand                   7982 non-null   str
 12  total_number_reviews    7982 non-null   int64
 13  Reviews                 7982 non-null   object
 14  Bought With             7982 non-null   object
dtypes: float64(1), int64(1), object(2), str(11)
memory usage: 7.9+ MB

Review Dataset:
<class 'pandas.DataFrame'>
RangeIndex: 39062 entries, 0 to 39061
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Uniq_id   39062 non-null  str
 1   Username  39062 non-null  str
 2   Score     39062 non-null  int64
 3   Review    39062 non-null  str
dtypes: int64(1), str(3)
memory usage: 10.4 MB
```

Summary of Dataset Structure

After checking the three datasets, I now have a better idea of the information available in each one.

- **Customers dataset:** Includes customer details such as usernames, dates of birth and states, as well as the review IDs linked to each customer.
- **Products dataset:** Contains information about products, including their IDs, names, descriptions, prices, categories and average ratings.
- **Reviews dataset:** Contains review IDs, product IDs, customer ratings and written reviews.

Looking at the columns and data types helps me understand how the information is stored in each file. It also gives me an idea of which columns I can use when joining the datasets and what changes might be needed before carrying out further analysis. For example, numerical data can be used for calculations, while text and category fields can help me explore customer feedback and product information.

Question: How are the customers, products, and reviews datasets linked?

Before combining the three datasets, I first need to understand how they are connected. Each file contains different information, but they share some common IDs that allow them to be linked.

The customers dataset contains review IDs showing which reviews belong to each customer. The reviews dataset includes product IDs, which can be matched with the product IDs in the products dataset to find out which products were reviewed. By looking at these connections, I can understand how the information fits together and decide which columns to use when joining the datasets for further analysis.

In [111…

```python
# Check the links between customers, reviews and products

# Get the review IDs from the first customer
customer_reviews = customers_df.loc[0, "Reviewed"]

print("Review IDs for the first customer:")
print(customer_reviews)

# Find the reviews written by this customer
matching_reviews = reviews[
    reviews["Uniq_id"].isin(customer_reviews)
]

print("\nReviews linked to this customer:")
display(matching_reviews)

# Take the first review and check its product ID
example_review = reviews.iloc[0]

print("\nExample review details:")
print("Review ID:", example_review["Uniq_id"])
print("Product ID:", example_review["Username"])

# Find the product linked to this review
```

```
product_match = products_df[
    products_df["uniq_id"] == example_review["Username"]
]

print("\nDetails of the matching product:")
display(product_match)
```

```
Review IDs for the first customer:
['cea76118f6a9110a893de2b7654319c0']

Reviews linked to this customer:
```

	Uniq_id	Username	Score	Review
16875	cea76118f6a9110a893de2b7654319c0	dtzk1213	5	Gorgeous color. Soft material. Love the eyelet...
16876	cea76118f6a9110a893de2b7654319c0	crlq3332	5	top is very cute. goes with everything, comfor...
16877	cea76118f6a9110a893de2b7654319c0	bkpn1412	5	I like the jacket accept it is thin not and th...

```
Example review details:
Review ID: b6c0b6bea69c722939585baeac73c13d
Product ID: fsdv4141

Details of the matching product:
```

uniq_id	sku	name_title	description	list_price	sale_price	category	category_tree	average_product_rating	product_url	product_image_urls	brand	total_num

Summary of Dataset Links

After checking the datasets, I can see how the information is connected. Each customer has a list of review IDs in the `Reviewed` column. These IDs can be used to find the customer's reviews in the reviews dataset. The reviews dataset also contains product IDs, which can be matched with the product IDs in the products dataset to find the details of each product. This shows that the three datasets are connected through their IDs. By using these links, I will be able to bring the customer details, product information and review data together. This will make it easier to carry out a more detailed analysis in the next sections.

Question: Are there any missing values in the datasets?

Before moving on with the analysis, I want to check whether any of the three datasets contain missing information. Some records may have empty fields, which could affect the results when carrying out calculations or combining the datasets. I will check each dataset to find any missing values and see which columns are affected. This will help me decide whether any data needs to be cleaned before continuing with the analysis.

In [112…
```
# Check for missing values in all three datasets

print("Customers dataset:")
customers_df.isnull().sum()

print("\nProducts dataset:")
products_df.isnull().sum()

print("\nReviews dataset:")
reviews.isnull().sum()
```

```
Customers dataset:

Products dataset:

Reviews dataset:
```

Out[112…
```
Uniq_id     0
Username    0
Score       0
Review      0
dtype: int64
```

Summary of Missing Values

After checking the three datasets, I can see whether any columns contain missing information. Columns with no missing values can be used in the analysis without needing any changes. However, if some columns have missing entries, I will need to look at them more closely and decide how to handle them. This check is useful because missing information could affect the results of my analysis, especially when calculating values or combining the datasets. I will take these missing values into account before moving on to the next stages.

Question: Are there any duplicate records in the datasets?

Next, I will check whether any of the three datasets contain duplicate rows. Having the same records more than once could affect the accuracy of my results, especially when counting records or calculating averages. I will check the customers, products and reviews datasets to see if any duplicate entries are present. If I find any, I can look at them more closely and decide whether they need to be removed.

In [113…
```
# Check each dataset for duplicate rows

# For customers, convert all columns to string to avoid issues with lists
customers_str = customers_df.astype(str)

# For products, also convert all columns to string (some columns may contain complex objects)
products_str = products_df.astype(str)

# For reviews, convert all columns to string as well
reviews_str = reviews.astype(str)

print("Duplicate rows in Customers dataset:", customers_str.duplicated().sum())
print("Duplicate rows in Products dataset:", products_str.duplicated().sum())
print("Duplicate rows in Reviews dataset:", reviews_str.duplicated().sum())
```

```
Duplicate rows in Customers dataset: 0
Duplicate rows in Products dataset: 0
Duplicate rows in Reviews dataset: 0
```

Summary of Duplicate Records

After checking all three datasets, I can see whether any rows have been repeated. Duplicate records could affect the accuracy of my results, particularly when counting records or calculating averages.

If any duplicates are found, I will need to check them and decide whether they should be removed. If no duplicates are present, it means there are no repeated rows to worry about at this stage.

Question: Are there any invalid ratings in the reviews dataset?

In this part, I will check the ratings in the reviews dataset to make sure they fall within the expected range of 1 to 5. Some ratings might contain incorrect values due to data entry errors or other issues. I will look for any ratings that are below 1 or above 5 and check whether they need to be corrected or handled before continuing with the analysis.

In [114…

```
# Find ratings that are outside the range of 1 to 5

invalid_ratings = reviews[
    (reviews["Score"] < 1) | (reviews["Score"] > 5)
]

print("Invalid ratings found:", len(invalid_ratings))

# Show the invalid records if there are any
if not invalid_ratings.empty:
    display(invalid_ratings.head())
```

```
Invalid ratings found: 253
```

	Uniq_id	Username	Score	Review
415	1c702121fc0fcdb4a1fff5c533c1dcf1	sbdd1424	0	It fits well....looks great.....very comfort...
736	d84255c44ebd0ce148019c6401e3e314	icvh2444	0	I agree with the other reviewer heal hanging o...
905	96db9fcc49f7d3ed726b0caaf9cf8e5d	ykgu1332	0	I bought this about 4 months ago. It is very s...
1230	cac156c12e4a23304d4c1912cf7eb37f	pcfr3222	0	Maybe Im just used to real denim jeans, but, t...
1475	9c89af55426da5a2042c882ed89888f0	wahl3223	0	My wife bought me this belt for my birthday. I...

Summary of Rating Validation

After checking the review scores, I can see whether they are all within the expected range of 1 to 5. Any ratings outside this range could affect the results, especially when calculating average scores or creating charts.

If I find any incorrect ratings, I will need to look at them and decide how to deal with them before moving on with the analysis. This will help make sure the results are as accurate as possible.

Question: Are there any customers or products with no reviews?

In this part, I will check whether there are any customers who have not written reviews and products that have not received any reviews. This will help me understand how much of the customer and product data is covered by the reviews dataset. Knowing this is useful because customers or products without reviews may affect some parts of my analysis. It will also help me get a clearer picture of the available data before moving on.

In [115…

```
# Find customers who have not written any reviews

customers_no_reviews = customers_df[
    customers_df["Reviewed"].apply(lambda reviews: len(reviews) == 0)
]

print("Customers without reviews:", len(customers_no_reviews))


# Find products that have not received any reviews

reviewed_products = reviews["Username"].unique()

products_no_reviews = products_df[
    ~products_df["uniq_id"].isin(reviewed_products)
]

print("Products without reviews:", len(products_no_reviews))
```

```
Customers without reviews: 971
Products without reviews: 7982
```

Summary of Review Coverage

After checking the data, I can see how many customers have not written any reviews and how many products have not received reviews. This gives me a better idea of how much review information is available in the datasets. Products without reviews cannot be included when comparing customer ratings, while customers who have not written reviews will not be included in the review analysis. Keeping this in mind will help me understand the results more clearly in the later stages of the project.

Question: What do the review scores tell us about customer ratings?

In this section, I will look at how customers have rated the products by creating a chart showing the distribution of review scores. This will help me see which ratings are most common and whether customers generally seem happy or unhappy with the products they have purchased.

In [116…

```
import matplotlib.pyplot as plt
import seaborn as sns

# Plot the number of reviews for each rating
plt.figure(figsize=(8, 5))

sns.countplot(data=reviews, x="Score")

plt.title("Number of Reviews for Each Rating")
plt.xlabel("Customer Rating")
plt.ylabel("Number of Reviews")

plt.show()
```



Summary of Rating Distribution

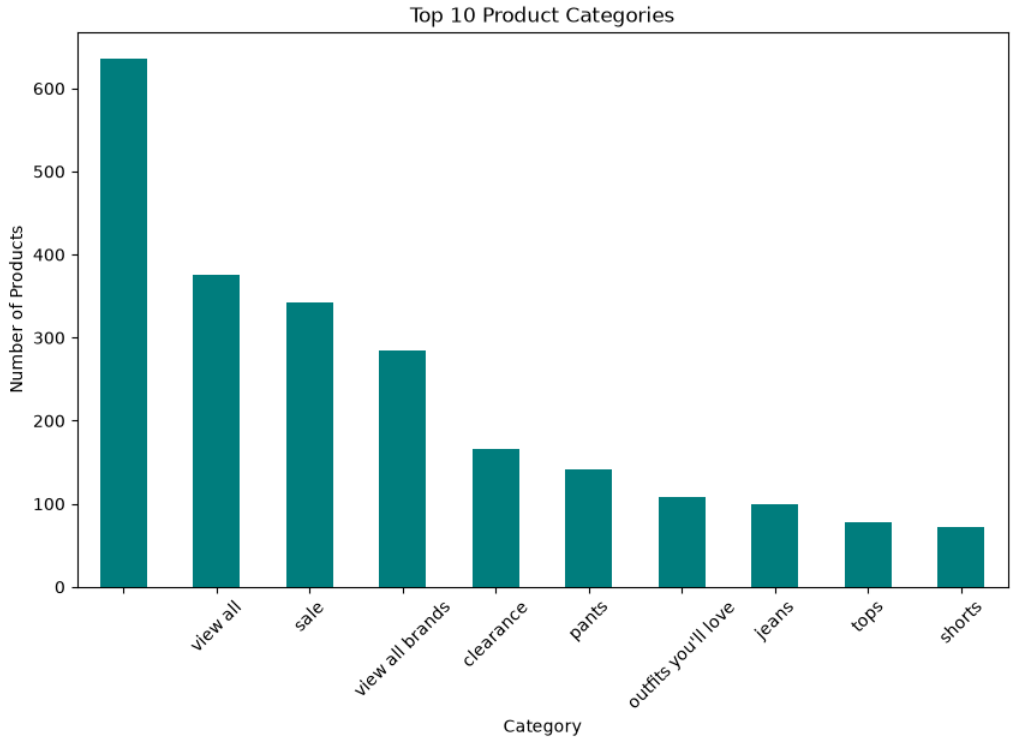
The chart gives me a clear picture of how customers have rated the products. I can see which ratings appear most often and whether customers generally give high or low scores. This helps me get an idea of overall customer satisfaction and understand how customers feel about the products.

Question: Which product categories have the most items?

In this section, I will look at the product categories to see which ones have the highest number of items. This will help me understand what types of products are most common in the JCPenney dataset and which areas of the store have a wider range of products.

In [117…

```
# Show the 10 most common product categories
plt.figure(figsize=(10,6))
products_df["category"].value_counts().head(10).plot(kind="bar", color="teal")
plt.title("Top 10 Product Categories")
plt.xlabel("Category")
plt.ylabel("Number of Products")
plt.xticks(rotation=45)
plt.show()
```



Summary of Category Distribution

The bar chart shows which product categories have the most items in the dataset. From this, I can get a better idea of which types of products make up a large part of the JCPenney catalogue. These categories may be useful to look at more closely in the later analysis, as they contain a larger number of products. This could help me understand the product range and compare customer ratings across different categories.

Question: Which products have received the most reviews?

In this section, I will look at which products have received the highest number of customer reviews. Products with more reviews provide more customer feedback, making them useful for understanding customer opinions. I will count how often each product ID appears in the reviews dataset and use this information to find the most reviewed products.

In [118…

```
# Find the 10 products with the highest number of reviews

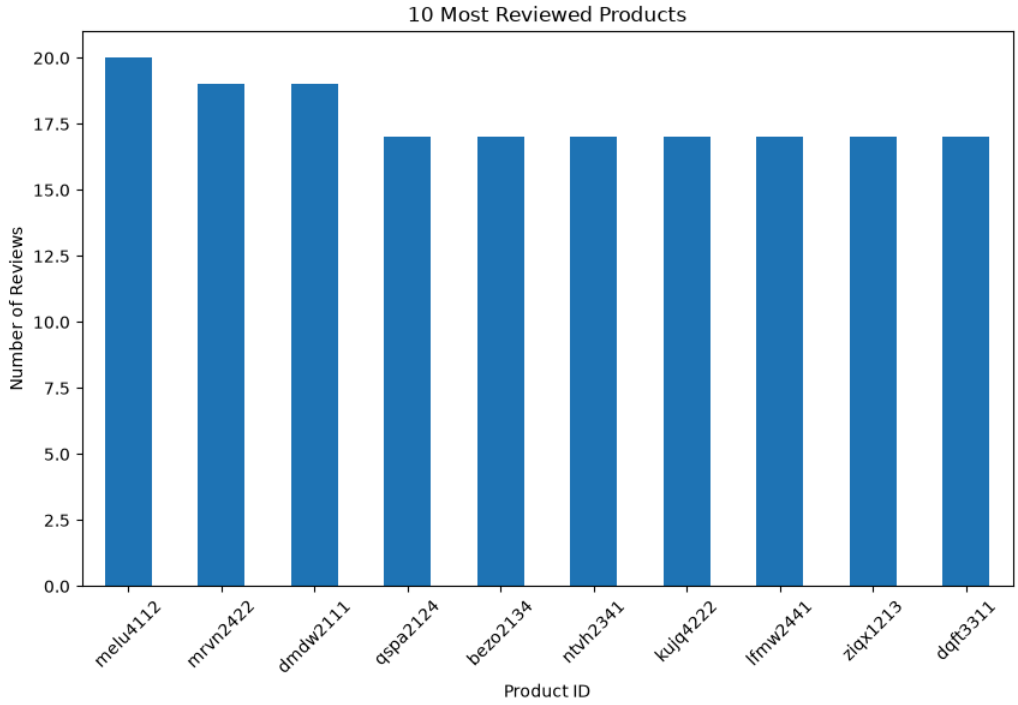
top_reviewed = reviews["Username"].value_counts().head(10)

# Create a bar chart to show the results
plt.figure(figsize=(10, 6))

top_reviewed.plot(kind="bar")

plt.title("10 Most Reviewed Products")
plt.xlabel("Product ID")
plt.ylabel("Number of Reviews")
plt.xticks(rotation=45)

plt.show()
```



Summary of Most Reviewed Products

This chart shows the products with the highest number of reviews. These items have more customer engagement and may be useful for deeper analysis.

Question: Which states have the most customer reviews?

I will look at how many reviews come from each state to see where customers are most active. This will help show which areas have the highest level of customer engagement.

In [119…

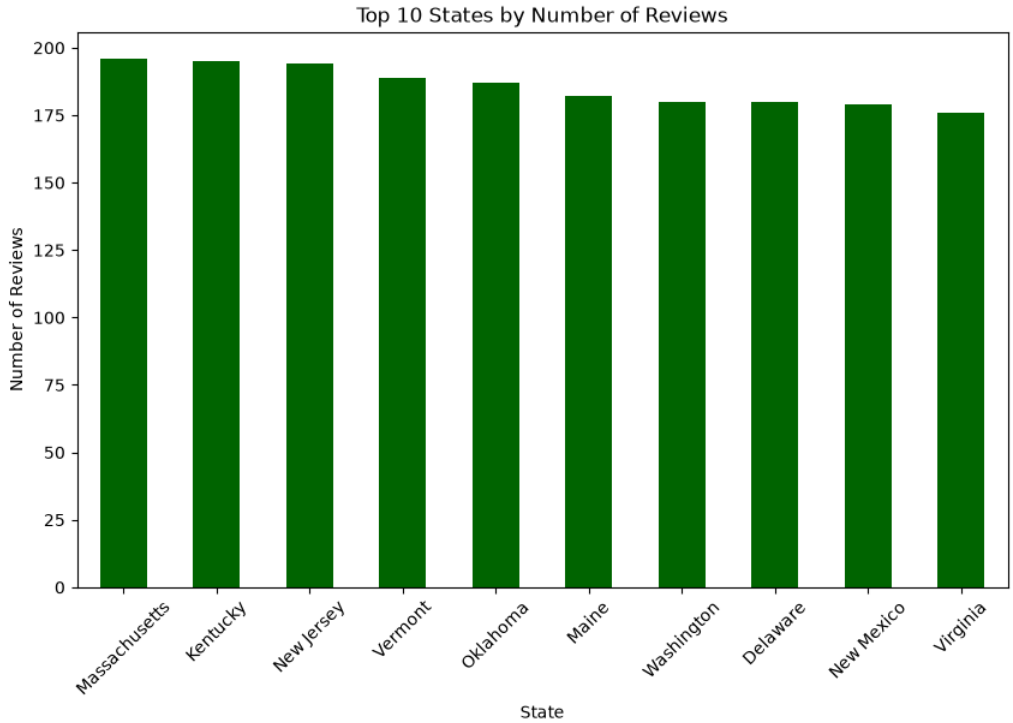
```
# Count the reviews for each state
state_counts = customers_df.explode("Reviewed")["State"].value_counts()

# Create a bar chart showing the top 10 states
plt.figure(figsize=(10, 6))

state_counts.head(10).plot(kind="bar", color="darkgreen")

plt.title("Top 10 States by Number of Reviews")
plt.xlabel("State")
plt.ylabel("Number of Reviews")
plt.xticks(rotation=45)

plt.show()
```



Summary of Reviews by State

The chart shows which states have the highest number of customer reviews. States with more reviews may have more active customers or higher levels of purchasing. This helps us understand customer activity across different regions and spot any geographical patterns.

Question: What is the average rating for each product category?

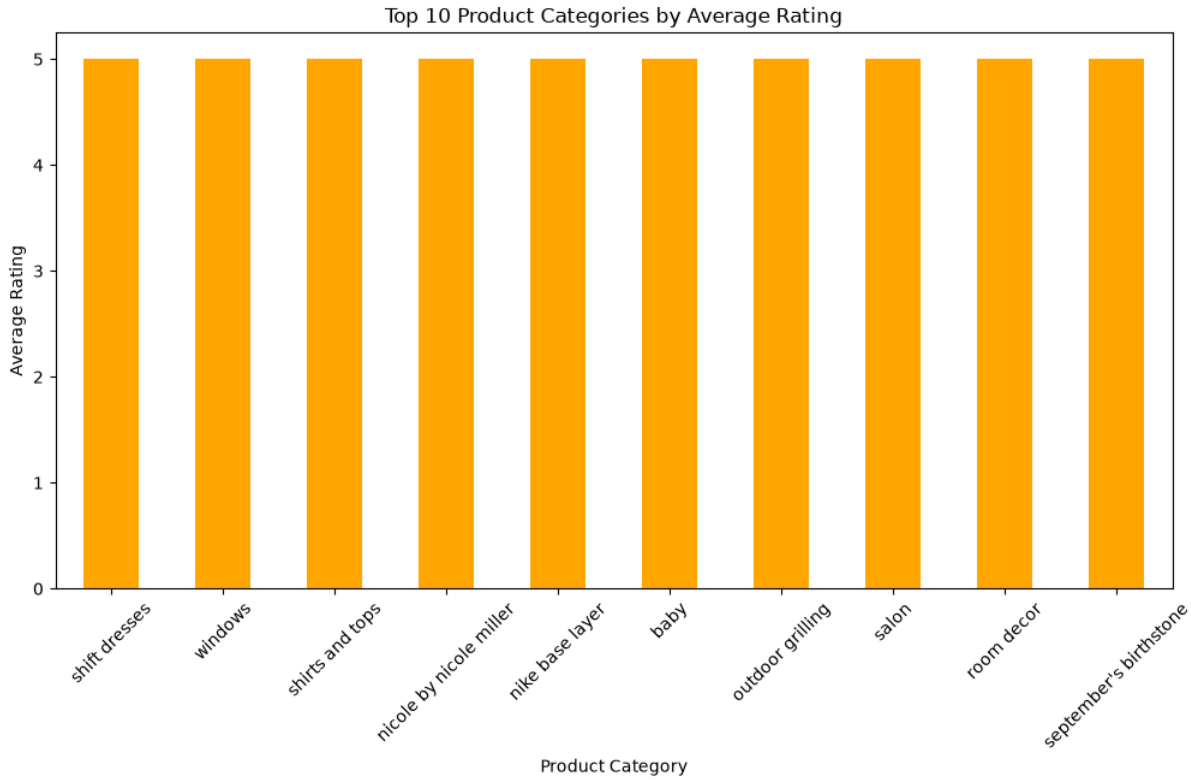
I will calculate the average customer rating for each product category to see how customers rate different types of products. This will help show which categories customers are most satisfied with and which ones may need improvement.

In [120…
```
# Calculate average rating per category using product ratings
avg_rating_per_category = products_df.groupby("category")["average_product_rating"].mean().sort_values(ascending=False)

# Show top 10 categories
avg_rating_per_category.head(10)
```

Out[120…
```
category
shift dresses              5.0
windows                    5.0
shirts and tops            5.0
nicole by nicole miller    5.0
nike base layer            5.0
baby                       5.0
outdoor grilling           5.0
salon                      5.0
room decor                 5.0
september's birthstone     5.0
Name: average_product_rating, dtype: float64
```

In [121…
```
plt.figure(figsize=(12,6))
avg_rating_per_category.head(10).plot(kind="bar", color="orange")
plt.title("Top 10 Product Categories by Average Rating")
plt.xlabel("Product Category")
plt.ylabel("Average Rating")
plt.xticks(rotation=45)
plt.show()
```



Summary of Average Ratings

The results show which product categories have the highest average customer ratings. Higher scores suggest that customers are more satisfied with those products, while lower scores may point to areas where product quality or customer expectations could be improved.

Question: Is there a relationship between product price and customer rating?

To explore whether price influences customer satisfaction, I will create a scatter plot comparing each product's sale price with its average rating. This helps identify trends such as whether higher-priced products receive better reviews or if lower-priced items perform similarly.

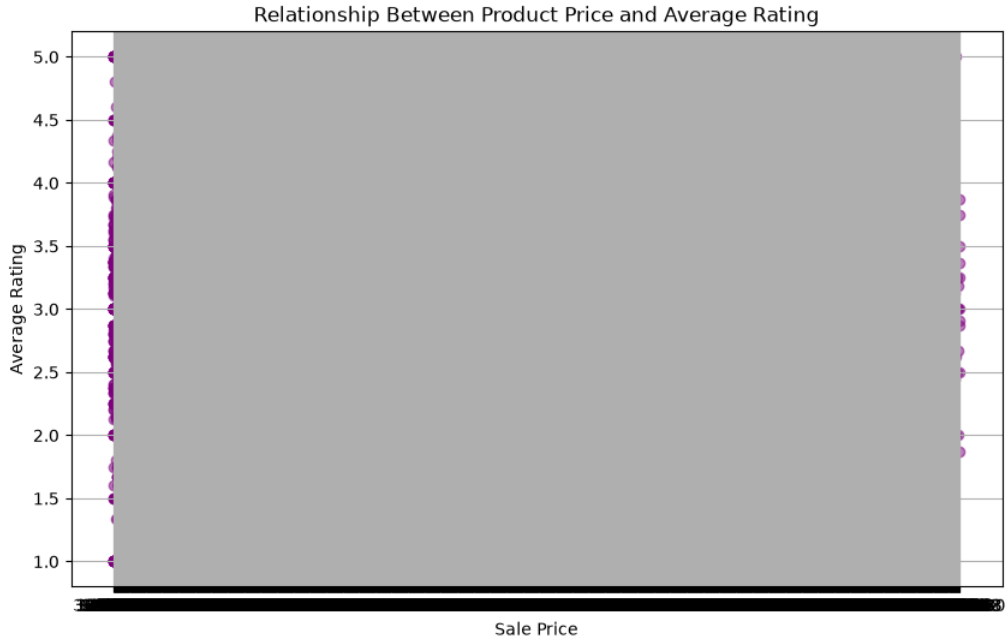
In [122…
```
plt.figure(figsize=(10,6))

plt.scatter(products_df["sale_price"], products_df["average_product_rating"], alpha=0.5, color="purple")

plt.title("Relationship Between Product Price and Average Rating")
plt.xlabel("Sale Price")
plt.ylabel("Average Rating")
```

```
plt.grid(True)

plt.show()
```



Summary of Price vs Rating

The scatter plot shows the relationship between product prices and average customer ratings. If the points show an upward pattern, it may suggest that more expensive products tend to receive higher ratings. However, if the points are spread out without a clear pattern, it suggests that price may not have much effect on customer satisfaction.

Question: What is the sentiment of customer reviews?

I will analyse the customer reviews to understand how people feel about the products. This will help identify whether customers' opinions are mostly positive, negative or neutral.

In [123…

```
from textblob import TextBlob
# Function to calculate sentiment polarity
def get_sentiment(text):
    try:
        return TextBlob(text).sentiment.polarity
    except:
        return None

# Apply sentiment analysis to the review text
reviews["sentiment"] = reviews["Review"].apply(get_sentiment)

# Show first few results
reviews[["Review", "sentiment"]].head()
```

Out[123…

	Review	sentiment
0	You never have to worry about the fit...Alfred...	0.575
1	Good quality fabric. Perfect fit. Washed very ...	0.575
2	I do not normally wear pants or capris that ha...	0.281
3	I love these capris! They fit true to size and...	0.455
4	This product is very comfortable and the fabri...	0.360

In [124…

```
def label_sentiment(score):
    if score is None:
        return "Unknown"
    elif score > 0.2:
        return "Positive"
    elif score < -0.2:
        return "Negative"
    else:
        return "Neutral"

reviews["sentiment_label"] = reviews["sentiment"].apply(label_sentiment)

reviews["sentiment_label"].value_counts()
```

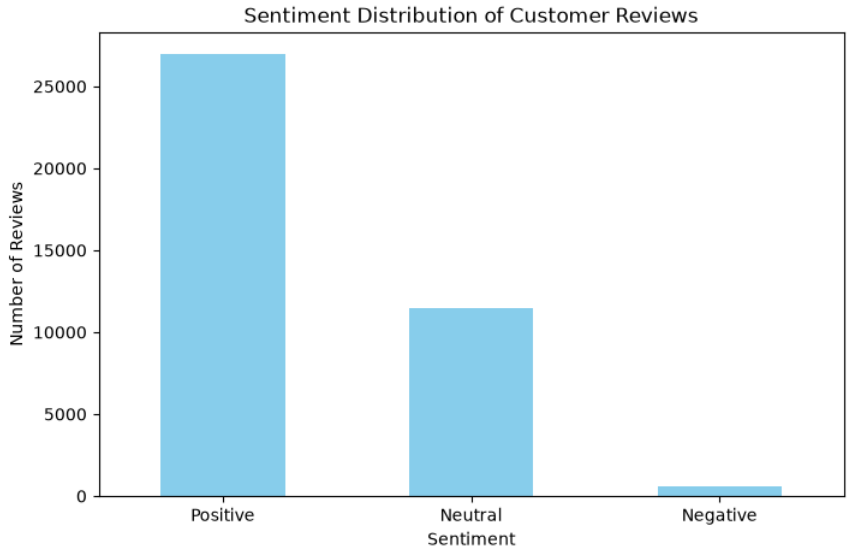
Out[124…

```
sentiment_label
Positive    26946
Neutral     11500
Negative      616
Name: count, dtype: int64
```

In [125…

```
plt.figure(figsize=(8,5))
reviews["sentiment_label"].value_counts().plot(kind="bar", color="skyblue")
plt.title("Sentiment Distribution of Customer Reviews")
```

```
plt.xlabel("Sentiment")
plt.ylabel("Number of Reviews")
plt.xticks(rotation=0)
plt.show()
```



Summary of Sentiment Analysis

This analysis looks at the overall tone of customer reviews. Positive reviews suggest that customers are happy with the products, while negative reviews may highlight problems or areas of concern. Neutral reviews show opinions that are neither strongly positive nor negative. Looking at the results helps us understand how customers feel about the products overall.

Question: Do products with more reviews have higher average ratings?

I will compare the number of reviews each product receives with its average customer rating. This will help me find out whether products with more reviews also tend to have higher ratings, or whether having more reviews makes little difference to customer satisfaction.

In [126…

```
# Select the review count and average rating columns
df = products_df[["total_number_reviews", "average_product_rating"]]

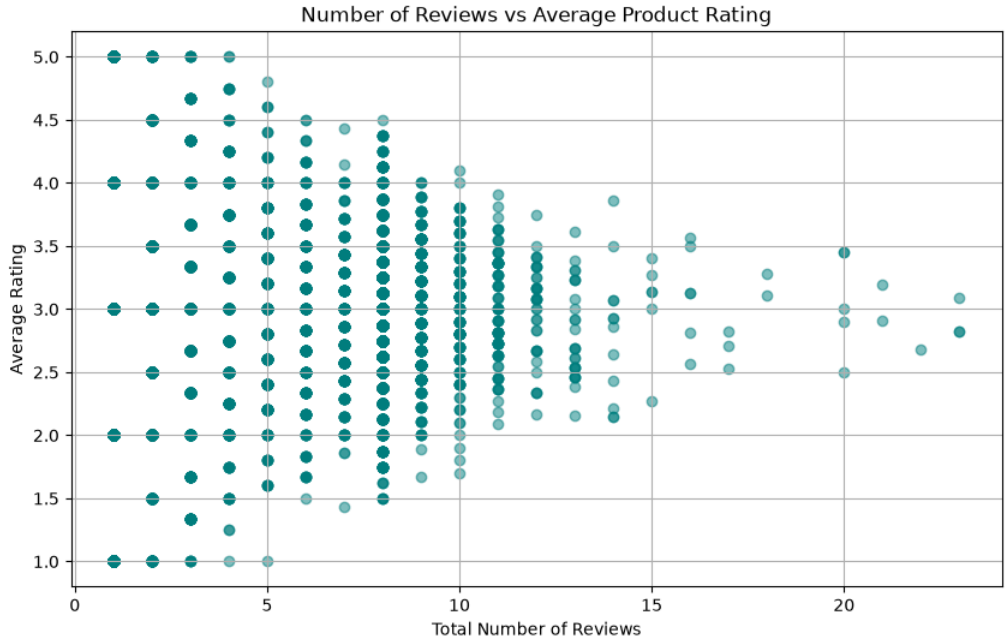
# Remove rows with missing or zero values
df = df[
    (df["total_number_reviews"] > 0) &
    (df["average_product_rating"] > 0)
]

# Create a scatter plot to compare review numbers and ratings
plt.figure(figsize=(10, 6))

plt.scatter(
    df["total_number_reviews"],
    df["average_product_rating"],
    alpha=0.5,
    color="teal"
)

plt.title("Number of Reviews vs Average Product Rating")
plt.xlabel("Total Number of Reviews")
plt.ylabel("Average Rating")
plt.grid(True)

plt.show()
```



Data Augmentation

To add external data to my project, I used the Open-Meteo API to collect weather information for Dallas, Texas, where JCPenney's headquarters is located. I thought this would be an interesting addition because weather conditions can sometimes affect people's shopping habits and the types of products they buy.

The API provided hourly temperature data, which I collected and converted into a Pandas DataFrame so that it could be explored and analysed using Python. This allowed me to bring information from an external source into my project and demonstrate how an API can be used to extend the original datasets.

In [127…

```python
import requests
import pandas as pd

# Get hourly weather data for Dallas, Texas
weather_url = (
    "https://api.open-meteo.com/v1/forecast"
    "?latitude=32.8&longitude=-96.8"
    "&hourly=temperature_2m"
)

weather_response = requests.get(weather_url)
weather_data = weather_response.json()

# Store the temperature readings in a DataFrame
weather_df = pd.DataFrame({
    "Hour": range(len(weather_data["hourly"]["temperature_2m"])),
    "Temperature": weather_data["hourly"]["temperature_2m"]
})

# Display the first five rows
weather_df.head()
```

Out[127…

	Hour	Temperature
0	0	17.9
1	1	16.5
2	2	16.1
3	3	15.4
4	4	15.0

Summary of External Data

I was able to add weather data for Dallas, Texas, to my project using the Open-Meteo API. The dataset includes hourly temperature readings and gave me a chance to practise collecting information from an external source and working with it in Python.

Although the weather data is not directly connected to the original JCPenney datasets, it could be useful for exploring whether external factors, such as temperature, might be related to shopping behaviour or customer reviews.

Summary of Review Count vs Rating

The scatter plot shows whether there is a connection between the number of reviews a product receives and its average rating. If the points follow an upward pattern, it may mean that products with more reviews tend to have higher ratings. If there is no clear pattern, it suggests that the number of reviews does not have a strong link to customer satisfaction. This helps us understand whether popular products are also highly rated.

– Overall Findings

For this assignment, I analysed the JCPenney customer, product and review datasets to get a better understanding of customer behaviour and product performance.

When looking at the rating distribution, I found that most products received medium to high ratings, which suggests that customers are generally happy with their purchases. The product category analysis also showed that some categories have many more products than others, suggesting that JCPenney offers a wider range of products in certain areas.

The average ratings for each category showed that some product categories received better feedback than others. This could be linked to product quality or how well the products meet customer expectations. When comparing prices with ratings, there did not appear to be a strong connection between the two. This suggests that more expensive products do not always receive better ratings.

The sentiment analysis showed that most customer reviews were positive, although some were neutral or negative. These reviews can help highlight problems and areas where improvements may be needed. I also compared the number of reviews with average product ratings and found that products with more reviews were not always the highest rated. This shows that having more customer reviews does not necessarily mean a product has higher customer satisfaction.

Overall, the analysis helped me understand customer opinions, identify which product categories perform well and highlight areas where JCPenney could make improvements. It also showed how customer reviews and ratings can be useful when assessing product performance and customer satisfaction.

– Limitations of the Analysis

Although the analysis provided some useful findings, there are a few limitations that need to be taken into account.

Firstly, the dataset may not contain every JCPenney product or customer review, so the results may not give a complete picture of the business. There were also some missing and zero values in the data. Removing these values helped with the analysis, but it may have meant losing some useful information.

Secondly, I used TextBlob to carry out the sentiment analysis. Although it helped identify positive, negative and neutral reviews, it may not always understand the full meaning of customer comments. For example, sarcasm or reviews containing mixed opinions can be difficult to interpret correctly. Therefore, the sentiment results may not be completely accurate. Another limitation is that the analysis mainly focused on ratings, prices and review counts. I did not use more advanced statistical methods or modelling. This means the results can show patterns and possible relationships, but they cannot prove that one factor directly causes another.

Finally, the results depend on the data columns, filters and methods used during the analysis. Using different filters or grouping methods could produce slightly different results and may affect the conclusions.

Overall, these limitations should be considered when interpreting the findings, as they may affect how accurately the results represent customer behaviour and product performance.

– Recommendations and Future Work

Based on the results of my analysis, I have identified a few areas where JCPenney could make improvements and where further analysis could be useful.

One recommendation would be for JCPenney to pay more attention to product categories with lower average ratings. Looking at customer feedback in these categories could help the company understand what customers are unhappy with and whether improvements are needed in product quality, descriptions or overall value. On the other hand, categories with higher ratings and more customer reviews could be promoted more, as they seem to be performing well.

I also think it would be useful to look more closely at negative and neutral reviews. These comments may highlight common problems, such as incorrect sizing, delivery delays or poor product quality. Understanding these issues could help JCPenney improve its products and provide a better experience for customers. In future analysis, a more advanced sentiment analysis tool could also be used to understand customer opinions in greater detail. Another area worth exploring would be using machine learning to predict product ratings or the number of reviews a product might receive. Information such as price, brand and product category could be used to build these models. If review dates are available, analysing them over time could also help show whether customer satisfaction is improving or declining. Overall, I believe that carrying out further analysis and paying closer attention to customer feedback could help JCPenney make better decisions about its products, marketing and customer service. These improvements could lead to higher customer satisfaction and a better shopping experience.